

CSE231 Operating System (Section A), Quiz 02 Monsoon 2025

Time allocated: 4pm – 4:20pm

Name	
Roll Number	

Instructions:

- This is a closed book and closed notes quiz. Please be aware of strict plagiarism policy.
- For questions requiring justification, please be as concise as possible. 2-3 sentences would be the ideal size of a justification. No extra pages will be provided.
- **NO EXTRA ANSWER SHEET WILL BE PROVIDED. USE THE GIVEN SPACE WISELY**

Question-1: Arrange the following events in the chronological order assuming that the parent process mentioned in event “d” is already running: a) orphan, b) fork, c) exit(0), d) parent terminating due to segmentation fault, e) loader invoked, and f) execv. Only provide the option numbers in correctly arranged order. No justifications needed. **[1 marks]**

Answer:

Note that it is mentioned that the parent participating in this set of events was already running. Hence, the loader cannot be the first event. It must happen after execv. Also, a child becomes orphan as soon as its parent dies. Hence, d→a must be in exact same sequence. Also, a parent cannot terminate before fork in this case. Also, as parent segfault it cannot call exit (i.e., it must have been called by forked process).

b → f → e → d → a → c

OR

b → f → d → a → e → c

OR

b → d → a → f → e → c

[No partial marking here]

Question-2: In the above-mentioned list of events, when arranged chronologically, a zombie process will also be created. True/False? Give a one-line justification. **[2 marks]**

Answer:

False [+1 mark]. A zombie process is created only if a child terminates before its parent and the parent has not yet called a wait or waitpid (i.e., if the child did not become an orphan) [+1 mark].

Question-3: A semaphore S has been allocated in shared memory and initialized with integer value X. S is shared among a group of ten processes. What should be the value of X such that at most two out of ten processes always remain unblocked while calling sem_wait? Give a one-line justification. **[2 marks]**

Answer:

X=2 [+1 mark]. Process blocks in sem_wait only when X become negative after a decrement. Hence, X=2 will allow up to two processes to see non-negative value of X after decrementing by one [+1 mark].

Question-4: What should be the exact size of integer array A when A is passed as a parameter to pipe syscall? The size should be chosen such that none of the array A's slots are wasted. One line justification. **[1.5 marks]**

Answer:

It must be of size 2 [+0.5 mark]. Pipe syscall fill up the provided integer array with the fd for its read and write end [+1 mark].

Question-5: Write pseudocode in 2-3 lines only (excluding the line count for braces) that demonstrates the creation of 10 processes using a **for** loop. Each newly created process must not create any further processes. Do not count the process executing your pseudocode (the original executable). No justifications needed. **[2 marks]**

Answer:

```
for(int i=0; i<10; i++) {           [+0.5 mark]
    if(fork() == 0) {               [+0.5 mark] (code must match here)
        exit(0); OR break; OR execv [+1 mark] (no need to check syscall parameter)
    }
}
```

Question-6: Signals are called the minimal form of IPC. True/False? Provide a one-line justification. **[1.5 mark]**

Answer:

True **[+0.5 mark]**. **Only an integer value could be communicated (minimal data)** **[+1 mark]**.